



Faculty of Science and Technology

Department of Applied Mathematics and Statistics

Customer Churn Prediction Using Random Forest Machine Learning Algorithms: A Case Study of KCI Management Consultants Zimbabwe

BY

Ishmael Mururami : R227365A

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Introduction

Context

It is more expensive to sign in a new client than keeping an existing one. Losing existing customers reduces revenue, weakens customer loyalty and increases the cost of acquiring new clients. In microfinance institutions, customer retention is important due to the dependence on long-term customer relationships and repeat borrowing behaviour.

Key Issues

- Increasing customer attrition
- Reduced customer lifetime value
- High customer replacement costs
- Limited predictive retention systems

Focus

- Analyze factors influencing churn
- Train a Random Forest model
- Predict customer exit behavior
- Deploy a prediction system

Why Random Forest?

Random Forest algorithm provides high predictive accuracy, handles non-linear customer behaviour patterns, reduces overfitting and produces interpretable feature importance measures useful for managerial decision making.

Background to the Study

Overview

Studies show that retaining existing customers is significantly less costly than acquiring new ones (Kotler & Keller, 2016). Consequently, organizations are increasingly adopting predictive analytics and machine learning techniques to improve customer retention strategies.

Microfinance

Microfinance institutions rely on long-term customer relationships, repeat borrowing and trust. High churn rates negatively affect profitability, customer lifetime value, and institutional sustainability.

Machine Learning

Recent studies indicate that ML models such as XGBoost, Random Forest and Neural Nets are best in churn prediction due to their ability to model nonlinear behaviour patterns (Huang et al., 2012).

Major Motivation

Most local customer churn studies focus on identifying churn determinants and predictive analytics without deploying practical decision-support systems. This study combines customer retention analytics, ML and web-based deployment to support customer retention at KCI Management Consultants.

Problem Statement

Challenges

- KCI Management Consultants faces the risk of losing customers through churn.
- Customer attrition reduces revenue, customer loyalty and long-term profitability.
- Existing retention approaches are largely reactive rather than predictive.
- High risk customers are difficult to identify before they exit the institution.
- Many customer churn studies focus more on exploratory analysis and prediction accuracy without deploying usable decision support systems.
- The absence of intelligent churn prediction platform for KCI limits data driven retention strategies.

Proposed Solution

This study aims to develop and deploy a Random Forest-based application to predict individual level customer churn and support retention decision making at KCI Management Consultants.

Research Questions

- ① What are the major factors influencing customer churn at KCI Management Consultants?
- ② How effectively can a Random Forest model predict customer churn?
- ③ How can a deployed web-based prediction system support customer retention?

Research Objectives

Main Aim

To develop a Random Forest-based customer churn prediction and decision support system for KCI Management Consultants.

Objectives

- 1 To identify the key determinants influencing customer churn at KCI Management Consultants.
- 2 To develop and train a Random Forest machine learning model for predicting customer churn.
- 3 To deploy the trained churn prediction model through a web-based application.

Expected Outcomes

Expected Outcomes

- Identification of major churn drivers
- Trained and saved Random Forest prediction model
- Churn prediction website application

Justification of the Study

Academic Contribution

The study contributes to the growing body of knowledge in customer churn analytics, machine learning and predictive modelling within the microfinance sector.

Institutional Significance

- Supports customer retention
- Reduces customer attrition risk
- Improves decision making processes

Scope of the Study

Geographical and Entity Scope

The study focuses on KCI Management Consultants, a microfinance institution operating in Zimbabwe.

Conceptual Scope

- Customer churn analytics
- Random Forest machine learning
- Predictive modelling
- Model Deployment

Parametric Scope

- Demographic variables
- Behavioural indicators
- Financial indicators
- Customer engagement variables

Temporal Scope

The study uses customer churn data collected from the institutional dataset provided for research.

Study Assumptions

- The dataset represents enough customer behaviour at KCI Management Consultants.
- Customer churn patterns can be explained using the selected predictor variables.
- The Random Forest algorithm is suitable for churn prediction using the available dataset.
- Customer behaviour remains relatively stable during the study period.

Previous Studies in Zimbabwe

- Revesai et al. (2022) studied customer churn in Zimbabwean banks using Logistic Regression, Decision Trees, Random Forest, XGBoost and Deep Learning models. Neural Networks(87% accuracy) outperformed all models followed by random forests(86.5% accuracy), but the study had limited feature coverage and lacked deployment of a practical system.
- Chiurunge et al. (2023) developed an Artificial Neural Network model for churn prediction: Case study of Econet Wireless Zimbabwe. The study focused mainly on predictive accuracy without interpretable feature analysis and deployment integration.
- Vhimisai et al. (2024) examined customer switching behaviour in the Zimbabwean Mobile Telecommunication Industry using explanatory statistical approaches. The study did not implement machine learning models for churn prediction.
- Ahmed et al. (2024) compared Logistic Regression, Random Forest, GBM and SVM in retail banking churn prediction and found ensemble methods highly effective all with more than 80% accuracy, although the study did not focus on deploying systems within microfinance institutions.

Why Random Forest?

- Handles nonlinear relationships
- Reduces overfitting
- Produces feature importance
- Maintains strong predictive accuracy
- Most importantly, apart from proven strong predictive accuracy which XGBoost and Neural Networks do, there are best when interpretation is important like in the case of customer churn.

Research Gaps

- Limited studies in Zimbabwean microfinances
- No customer churn decision support system at KCI Management Consultants
- Limited integration of model deployment

Research Design

The study adopts a quantitative predictive analytics research design based on supervised machine learning techniques. The design focuses on analyzing customer behavior patterns and predicting customer churn using historical institutional data.

Characteristics

- Quantitative research approach
- Predictive modelling
- Case study of KCI Management Consultants
- Supervised classification problem

Why This Design?

- Supports churn prediction
- Enables statistical analysis
- Handles structured customer data
- Supports deployment of ML systems

Data Source

The study will use secondary customer data obtained from KCI Management Consultants. The dataset contains customer demographic, financial and behavioral variables associated with churn behavior.

Sampling Design

- Secondary institutional dataset
- Complete-case customer records
- Binary churn classification
- Customers represented as observations

Target Variable

- Churn = 1 : Customer exited
- Churn = 0 : Customer retained
- Supervised learning target
- Basis for prediction modelling

Data Preprocessing

To improve data quality, reduce inconsistencies and enhance model performance before training the Random Forest algorithm.

Data Cleaning

- Missing value assessment
- Duplicate record removal
- Outlier detection
- Data consistency checks

Feature Preparation

- Encoding categorical variables
- Feature selection
- Train-test split
- Class imbalance assessment

Train-Test Split

The dataset will be divided into training and testing datasets using

80% Training Data ; 20% Testing Data

Random Forest Classifier

The study will employ the Random Forest algorithm to classify customers according to their probability of churn. Random Forest is ensemble learning algorithm constructs multiple decision trees and combines their predictions to improve classification accuracy.

Random Forest Prediction

$$\hat{Y} = \text{Majority Vote}(T_1, T_2, T_3, \dots, T_n)$$

Model Optimization

- Hyperparameter tuning
- Cross-validation
- Tree depth optimization
- Feature importance analysis

Evaluation Metrics

The performance of the Random Forest model will be evaluated using classification metrics derived from the confusion matrix.

Confusion Matrix

$$\begin{bmatrix} TP & FP \\ FN & TN \end{bmatrix}$$

Performance Metrics

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

ROC AUC

Deployment Framework

The trained Random Forest model will be deployed to enable real-time customer churn prediction

Development and Deployment

- Pandas and NumPy for data processing
- Scikit-learn for Random Forest training
- Pickle/Joblib for model serialization
- Flask for web applications

Deployment Workflow

Data → Preprocessing → Training

→ Serialized Model → Flask API

→ Web Application → Prediction

Limitations of the Study

Methodological Limitations

The study acknowledges several methodological limitations that may influence the generalization and interpretation of findings.

Data Limitations

- Dependence on secondary institutional data
- Possible class imbalance in churn records
- Limited variables available in dataset

Model Limitations

- Results limited to KCI dataset
- Customer behavior may change over time
- Model performance depends on data quality

Measures

The study will apply preprocessing techniques, validation procedures and performance evaluation metrics to improve model reliability and reduce analytical bias.

Definition of Key Terms

Operational Definitions

Customer Attrition The gradual loss of customers over time due to termination of business relationships with an institution.

Customer Retention Strategies and processes used by organizations to maintain long-term customer relationships and reduce customer loss.

Customer Churn The process in which a customer discontinues their relationship with a financial institution.

Random Forest An ensemble machine learning algorithm that combines multiple decision trees for classification and prediction.

Predictive Analytics The use of statistical and machine learning techniques to predict future outcomes using historical data.

Machine Learning A branch of artificial intelligence that enables systems to learn patterns from data and make predictions.

Web-Based Deployment Implementation of a trained machine learning model within an online application for real-time usage.

Summary

Overview

The study proposes the development of a Random Forest-based customer churn prediction system for KCI Management Consultants, Zimbabwe. The research focuses on identifying major determinants of customer churn, developing a predictive machine learning model, and deploying the trained model through a web-based application.

Areas Covered

- Background to customer churn
- Problem statement
- Research objectives
- Literature review
- Methodological framework

Final Remark

The study seeks to contribute toward data-driven customer retention management in Zimbabwean microfinance institutions through the integration of machine learning and deployable predictive systems.

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